

Comparative Analysis of Machine Learning Algorithms for Breast Cancer

Molecular Subtype Classification Using Gene Expression Profiles

Iqra Shaikh

Poona College of Arts, Commerce and Science

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Abstract

Breast cancer is one of the most common cancers worldwide and is classified into different molecular subtypes based on gene expression patterns. Since these subtypes differ in their biological characteristics and response to treatment, accurate classification is important for early diagnosis and personalized therapy. In this study, machine learning techniques were used to classify breast cancer molecular subtypes using gene expression data from the publicly available GSE45827 dataset obtained from the Gene Expression Omnibus (GEO) database. After preprocessing, the final dataset consisted of 141 tissue samples belonging to Basal, HER2-positive, Luminal A, Luminal B, and Normal breast tissue groups.

The gene expression data were pre-processed by removing non-tissue samples and extracting subtype labels from the sample metadata. ANOVA F-test was used to select the most informative genes, and Principal Component Analysis (PCA) was performed to visualize the distribution of the selected samples. Four machine learning models—Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost—were trained using an 80:20 train-test split. To obtain a more reliable estimate of model performance, 10-fold stratified cross-validation was also carried out. The models were evaluated using accuracy, confusion matrices, ROC curves, and AUC scores.

Among the evaluated models, Logistic Regression and SVM achieved a test accuracy of 100%, while Random Forest and XGBoost achieved accuracies of 96.55%. The cross-validation results showed consistent performance across all models, with mean accuracies ranging from 90.12% to 96.44%, indicating good generalization on unseen data. The selected genes were further analyzed using Gene Ontology (GO) and KEGG pathway enrichment analysis, which revealed their involvement in several biological processes and pathways associated with breast cancer.

Overall, this study demonstrates that machine learning models, combined with feature selection and functional enrichment analysis, can effectively classify breast cancer molecular subtypes using gene expression data. The results also highlight the usefulness of integrating computational and biological analyses to better understand the molecular mechanisms underlying breast cancer and support future research in precision medicine.

Keywords: *Breast Cancer, Bioinformatics, Gene Expression Analysis, Machine Learning, Classification, Precision Medicine*

[Introduction](#)

Breast cancer is one of the leading causes of cancer-related deaths among women worldwide. According to the World Health Organization (WHO), millions of new breast cancer cases are diagnosed every year, making early detection and accurate diagnosis essential for improving patient survival. Breast cancer is not a single disease but consists of several molecular subtypes, including Luminal A, Luminal B, HER2-positive, and Basal (Triple Negative). These subtypes differ in their genetic characteristics, disease progression, prognosis, and response to treatment. Therefore, identifying the correct molecular subtype is an important step in selecting the most effective treatment strategy for patients.

With the rapid growth of genomic technologies, gene expression profiling has become an important tool for studying the molecular characteristics of breast cancer. Microarray-based gene expression datasets allow researchers to measure the expression levels of thousands of genes simultaneously, providing valuable information for disease classification and biomarker discovery. However, these datasets are usually high-dimensional, containing thousands of gene expression features but relatively few patient samples. This makes traditional statistical analysis challenging and creates a need for computational methods that can efficiently analyze such complex data.

Machine learning has emerged as a powerful approach for analyzing high-dimensional biological datasets. By learning patterns from gene expression data, machine learning models can classify different breast cancer subtypes with high accuracy and help identify genes that contribute most to the prediction process. In recent years, several algorithms such as Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost have been widely used in cancer classification studies because of their ability to handle complex datasets and generate reliable predictions.

In addition to accurate classification, identifying biologically significant genes is equally important. Feature selection techniques help reduce the dimensionality of gene expression data by selecting only the most informative genes while removing redundant or less relevant features. These selected genes can then be further investigated using functional enrichment analysis, such as Gene Ontology (GO) and Kyoto Encyclopaedia of Genes and Genomes (KEGG) pathway analysis, to better understand the biological processes and pathways associated with breast cancer development.

In this study, the publicly available GSE45827 gene expression dataset from the Gene Expression Omnibus (GEO) database was used to classify breast cancer molecular subtypes. After preprocessing the dataset, non-tissue samples were removed, and gene expression data from 141 tissue samples representing Basal, HER2-positive, Luminal A, Luminal B, and Normal breast tissues were included in the analysis. ANOVA F-test was applied for feature selection, followed by Principal Component Analysis

(PCA) for data visualization. Four supervised machine learning algorithms—Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost—were trained and evaluated using both an 80:20 train-test split and 10-fold stratified cross-validation. Model performance was assessed using accuracy, confusion matrices, ROC curves, and AUC scores. Finally, the selected genes were analyzed using GO and KEGG enrichment analysis to explore their biological significance.

The main objective of this study was to compare the performance of different machine learning algorithms for breast cancer molecular subtype classification and to investigate the biological relevance of the selected genes. By combining machine learning techniques with functional enrichment analysis, this work demonstrates how computational methods can support breast cancer research and contribute to the growing field of precision medicine.

[Literature Review](#)

Breast cancer is one of the most widely studied cancers because of its high prevalence and significant impact on global health, and the complexity of its molecular characteristics. Over the years, researchers have shown that breast cancer is not a single disease but consists of different molecular subtypes, each with its own gene expression pattern, prognosis, and response to treatment. A landmark study by Perou et al. (2000) introduced the molecular classification of breast cancer into Luminal A, Luminal B, HER2-positive, and Basal-like subtypes. This classification has become an important foundation for breast cancer diagnosis and treatment planning.

The development of microarray technology has made it possible to measure the expression levels of thousands of genes in a single experiment. As a result, large-scale gene expression datasets have become publicly available through databases such as the Gene Expression Omnibus (GEO). These datasets have allowed researchers around the world to investigate the molecular basis of breast cancer

and develop computational methods for disease classification. The GSE45827 dataset used in this study is one of the publicly available datasets that contains gene expression profiles from different breast cancer subtypes and has been used in several genomic studies.

In recent years, machine learning has become an important tool for analyzing gene expression data. Since these datasets contain thousands of genes but relatively few samples, traditional statistical methods often face challenges in identifying meaningful patterns. Machine learning algorithms are well suited for this type of data because they can learn complex relationships between genes and classify samples with high accuracy. Algorithms such as Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost have been successfully applied in several cancer prediction studies and continue to show promising results.

Another important aspect of gene expression analysis is identifying the genes that contribute most to disease classification. Feature selection techniques help reduce the number of genes by selecting only the most informative ones, making the models more efficient and easier to interpret. In addition, functional enrichment analysis using Gene Ontology (GO) and KEGG pathways helps researchers understand the biological roles of these genes and the pathways in which they are involved. This provides biological meaning to the computational results rather than focusing only on prediction accuracy.

Although many previous studies have reported high classification performance using machine learning, a large number of them mainly focus on developing predictive models. Relatively fewer studies combine machine learning with biological interpretation of the selected genes. In this study, an attempt was made to address both aspects. Gene expression data from the GSE45827 dataset were pre-processed, informative genes were selected using the ANOVA F-test, and four machine learning algorithms—Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost—were evaluated using both an independent test set and 10-fold stratified cross-validation. The selected genes

were further analyzed using GO and KEGG enrichment analysis to better understand their biological significance in breast cancer. By combining predictive modeling with functional analysis, this study provides a more comprehensive approach to breast cancer molecular subtype classification.

Study	Dataset	Methodology	Main Findings
Perou et al. (2000)	Breast tumor samples	Gene expression profiling	Identified intrinsic molecular subtypes of breast cancer.
Gruosso et al. (2016)	GSE45827	Microarray gene expression analysis	Developed a publicly available dataset representing multiple breast cancer molecular subtypes.
Recent machine learning studies	GEO and TCGA datasets	SVM, Random Forest, XGBoost	Demonstrated that machine learning models can accurately classify breast cancer subtypes.
Present Study	GSE45827	ANOVA feature selection, PCA, Logistic Regression, SVM, Random Forest, XGBoost, GO and KEGG enrichment analysis	Compared multiple machine learning models and combined predictive analysis with biological interpretation of selected genes.

The reviewed studies demonstrate that machine learning has become an effective approach for breast cancer subtype classification. However, there is still scope to integrate predictive modeling with biological interpretation of the selected genes. The present study attempts to address this by combining

machine learning techniques with functional enrichment analysis using the GSE45827 gene expression dataset.

Materials and Methods

Dataset Description

The gene expression dataset used in this study was obtained from the publicly available Gene Expression Omnibus (GEO) database maintained by the National Center for Biotechnology Information (NCBI). The GSE45827 dataset was selected because it contains gene expression profiles from different breast cancer molecular subtypes and has been widely used in breast cancer research. The expression data were generated using the Affymetrix Human Genome U133 Plus 2.0 microarray platform (GPL570), which measures the expression levels of thousands of genes simultaneously.

The original dataset contained 155 samples representing different breast cancer subtypes and tissue sources. During preprocessing, non-tissue samples and entries without relevant subtype information were removed. The final dataset consisted of 141 breast tissue samples belonging to five classes: Basal, HER2-positive, Luminal A, Luminal B, and Normal breast tissue.

Data Preprocessing

The downloaded gene expression matrix and sample metadata were imported into Python for preprocessing. Sample subtype information was extracted from the metadata, and only tissue samples representing the selected breast cancer subtypes were included in the analysis. Samples with incomplete or irrelevant subtype annotations were excluded to improve the quality of the dataset.

Since gene expression datasets contain a large number of variables, no missing values were observed after preprocessing, and the data were organized into a feature matrix consisting of gene expression values and their corresponding subtype labels. This cleaned dataset was then used for feature selection and model development.

After preprocessing, the original GSE45827 dataset was filtered to remove non-tissue samples and entries without relevant molecular subtype information. The final dataset consisted of **141 breast tissue samples** representing five molecular subtypes: Basal (41 samples), HER2-positive (30 samples), Luminal B (30 samples), Luminal A (29 samples), and Normal breast tissue (11 samples). The processed dataset contained **29,873 gene expression features**, resulting in a high-dimensional dataset suitable for machine learning analysis. The relatively balanced distribution of samples across the major subtypes helps reduce classification bias and allows the models to learn subtype-specific gene expression patterns more effectively.

[Feature Selection](#)

Gene expression datasets usually contain tens of thousands of genes, many of which contribute little to the classification task. To reduce the dimensionality of the dataset and improve model performance, feature selection was performed using the ANOVA F-test. The ANOVA F-test ranks genes according to their ability to distinguish between different breast cancer subtypes.

To avoid information leakage, feature selection was performed using only the training data within each modeling pipeline before evaluating model performance. The highest-ranked genes were selected and used for model training, reducing computational complexity while retaining the most informative features.

[Principal Component Analysis \(PCA\)](#)

Principal Component Analysis (PCA) was performed to visualize the distribution of breast cancer samples based on the selected gene expression features. PCA reduces high-dimensional gene expression data into a small number of principal components while preserving most of the variation present in the dataset. The resulting scatter plot provided an overview of sample clustering and the separation among different breast cancer molecular subtypes.

[Machine Learning Models](#)

- Four supervised machine learning algorithms were selected for comparison in this study:
- Logistic Regression (LR)
- Support Vector Machine (SVM)
- Random Forest (RF)
- Extreme Gradient Boosting (XGBoost)

The dataset was divided into training and testing sets using an 80:20 stratified split to preserve the class distribution. Each model was trained using the selected gene expression features, and predictions were generated for the testing dataset.

To obtain a more reliable estimate of model performance, 10-fold stratified cross-validation was performed. The mean accuracy and standard deviation obtained from cross-validation were used to evaluate the stability and generalization ability of each model.

[Model Evaluation](#)

The performance of each machine learning model was evaluated using several commonly used classification metrics. Test accuracy was calculated to measure the percentage of correctly classified samples. In addition, confusion matrices were generated to examine the classification performance for each breast cancer subtype. Receiver Operating Characteristic (ROC) curves and the Area Under the Curve (AUC) were also used to evaluate the discriminative ability of the models.

Cross-validation results were included to assess the consistency of model performance across different subsets of the dataset and to reduce the possibility of overestimating predictive accuracy.

[Gene Annotation and Functional Enrichment Analysis](#)

The top-ranked probe IDs obtained after feature selection were converted into gene symbols using the Affymetrix GPL570 platform annotation file. This annotation step enabled the biological interpretation of the selected genes.

To investigate the biological significance of these genes, Gene Ontology (GO) Biological Process enrichment analysis and Kyoto Encyclopaedia of Genes and Genomes (KEGG) pathway enrichment analysis were performed using the GSEAPy Python package. The significantly enriched biological processes and pathways provided insight into the molecular functions and signaling pathways associated with breast cancer subtype classification.

Results

Model Performance Evaluation

Dataset Summary

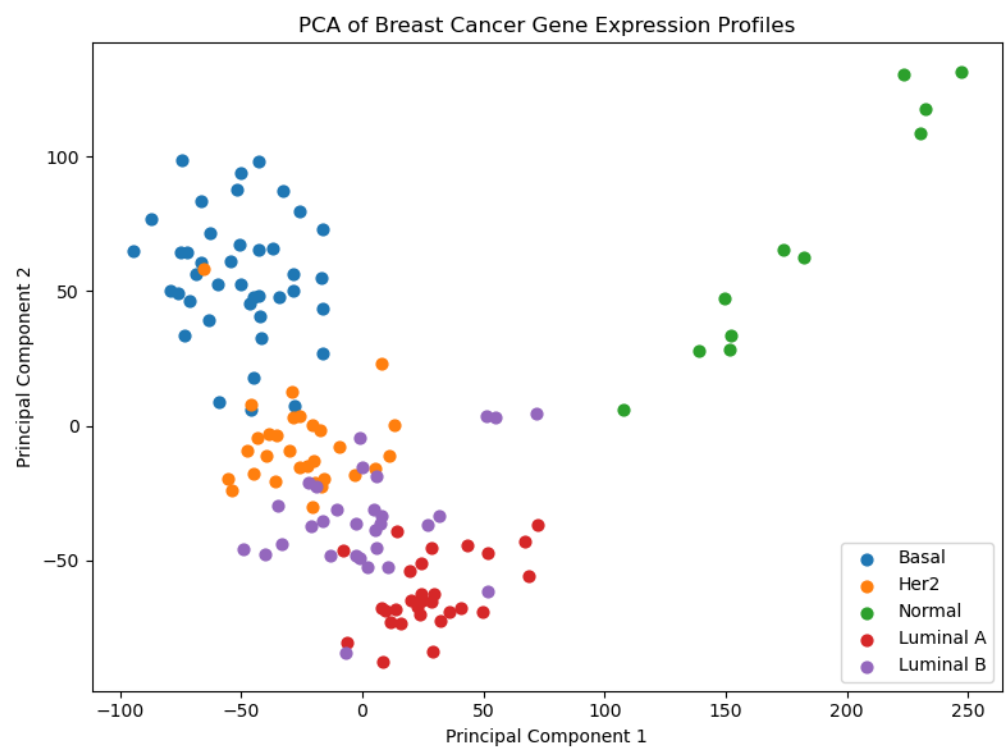
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Principal Component Analysis (PCA)

Principal Component Analysis (PCA) was performed to visualize the distribution of breast cancer samples after preprocessing. The PCA plot revealed that samples belonging to different molecular subtypes formed distinguishable clusters, indicating that the gene expression profiles captured meaningful biological differences among the subtypes. Basal and HER2-positive samples appeared more clearly separated from the other groups, whereas some overlap was observed between Luminal A and Luminal B samples. This overlap is biologically expected because both luminal subtypes share hormone receptor-

related molecular characteristics despite differences in their clinical behavior and prognosis. Overall, the PCA results suggest that the selected gene expression data contain sufficient variation to support accurate molecular subtype classification while also reflecting the biological complexity of breast cancer.

Figure 1. Principal Component Analysis (PCA) of the preprocessed GSE45827 dataset. Samples belonging to different molecular subtypes form distinguishable clusters, although partial overlap is observed between Luminal A and Luminal B, reflecting similarities in their underlying gene expression patterns.



Performance of Machine Learning Models

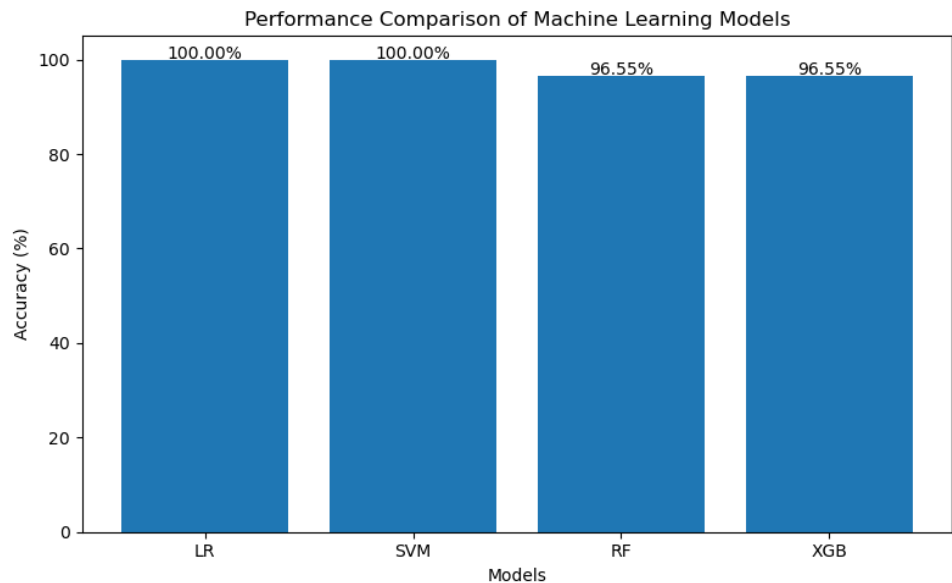
Four supervised machine learning algorithms were evaluated for breast cancer molecular subtype classification. The models were trained using the selected gene expression features and tested using an 80:20 stratified train-test split.

Table 1 summarizes the classification accuracy achieved by each model.

Table 1. Classification Performance of Machine Learning Models

Model	Test Accuracy (%)
Logistic Regression	100.00
Support Vector Machine	100.00
Random Forest	96.55
XGBoost	96.55

Among the evaluated models, Logistic Regression and Support Vector Machine achieved a test accuracy of 100%, correctly classifying every sample in the independent test set. Random Forest and XGBoost also demonstrated strong predictive performance, each achieving an accuracy of 96.55%. Although the difference in accuracy between the models was relatively small, the results indicate that both linear and non-linear machine learning algorithms can effectively distinguish breast cancer molecular subtypes when informative gene expression features are selected. These findings demonstrate the suitability of machine learning techniques for analyzing high-dimensional genomic datasets.



Cross-Validation Performance

To evaluate the robustness and generalization ability of the models, 10-fold stratified cross-validation was performed. The results showed that the Support Vector Machine achieved the highest average cross-validation accuracy (96.44%) with the lowest standard deviation, indicating stable performance across different data partitions. Logistic Regression also demonstrated strong and consistent performance, while Random Forest and XGBoost exhibited slightly greater variability. These findings suggest that although Logistic Regression and SVM achieved identical test accuracy, SVM generalized more consistently to unseen data and therefore appears to be the most reliable model for this dataset.

Table 2. Ten-Fold Cross-Validation Results

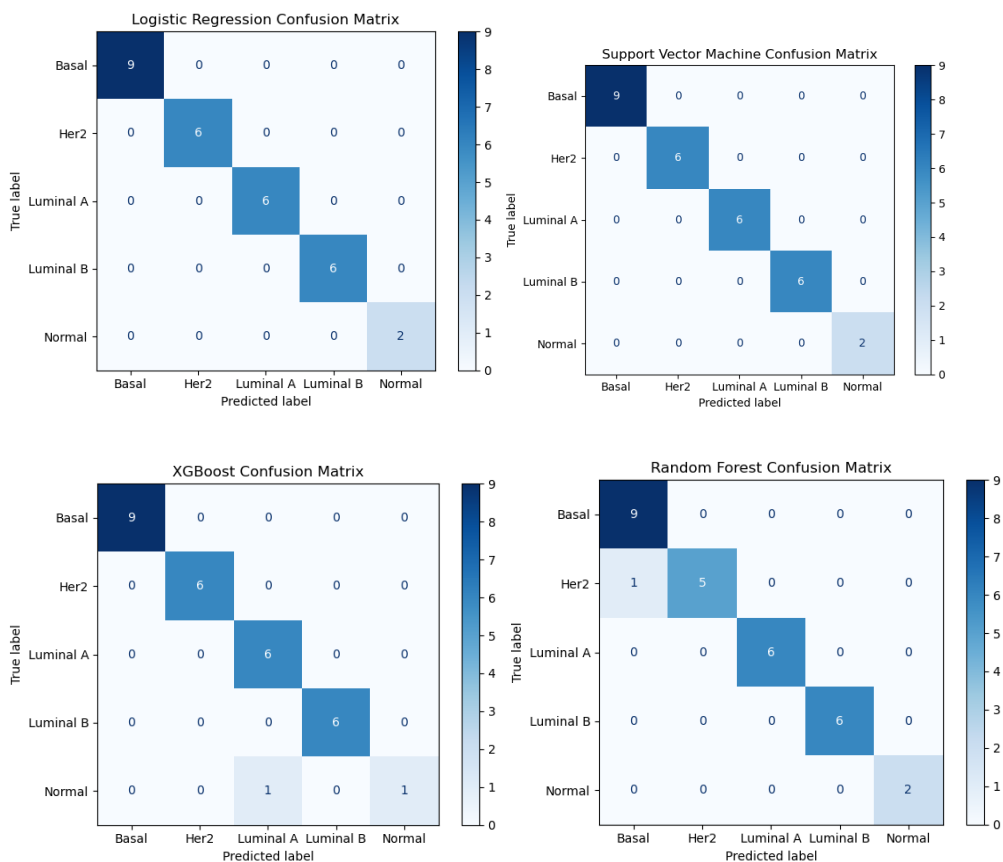
MODEL	MEAN ACCURACY	STANDARD DEVIATION
LOGISTIC REGRESSION	94.62%	1.86%
SUPPORT VECTOR MACHINE	96.44%	1.78%
RANDOM FOREST	94.62%	4.47%
XGBOOST	90.12%	5.22%

Although Logistic Regression and SVM achieved perfect test accuracy, the cross-validation results provide a more realistic estimate of model performance on unseen data. SVM achieved the highest average cross-validation accuracy (96.44%) with the lowest variation, indicating stable and consistent performance across different data splits.

Confusion Matrix Analysis

Confusion matrices were generated to examine the classification performance of each model across individual breast cancer subtypes. Logistic Regression and Support Vector Machine correctly classified all testing samples without any observed misclassification. In comparison, Random Forest and XGBoost produced only a small number of classification errors involving closely related subtypes. These results indicate that all four models were able to distinguish the molecular subtypes with high accuracy, while Logistic Regression and SVM demonstrated the strongest classification performance.

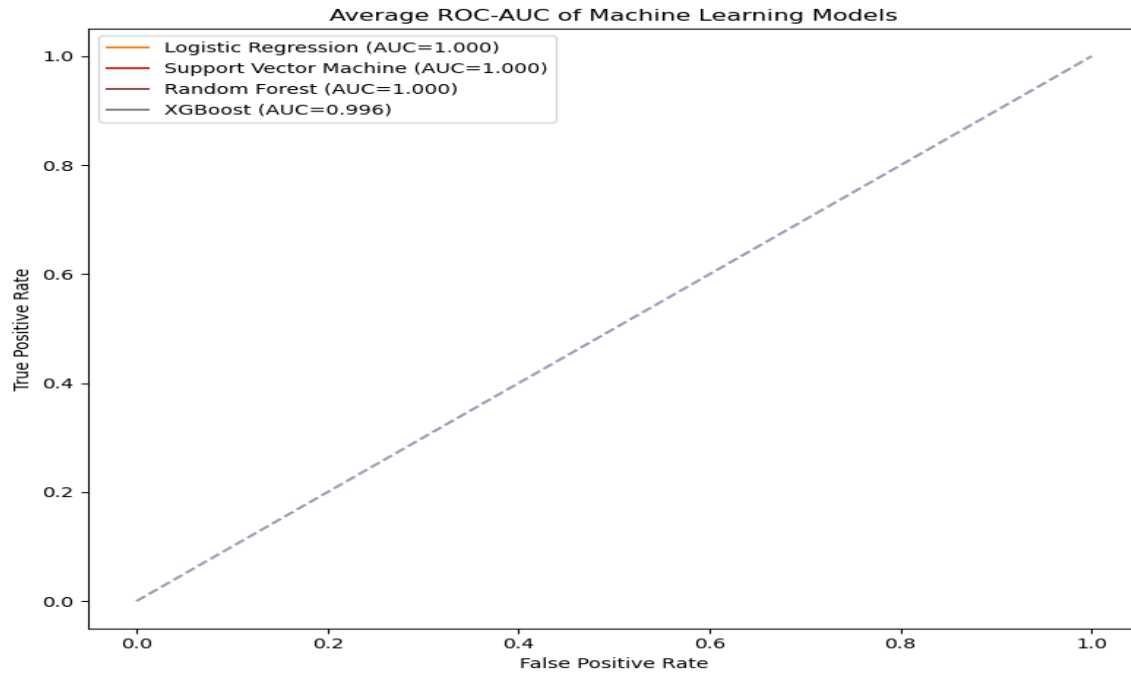
Figure 2. Confusion matrices of Logistic Regression, Support Vector Machine, Random Forest, and XGBoost models. Logistic Regression and SVM achieved perfect classification, whereas Random Forest and XGBoost produced only a few misclassified samples.



Receiver Operating Characteristic (ROC) Analysis

Receiver Operating Characteristic (ROC) curves were generated to evaluate the discriminative ability of the classification models. All four algorithms achieved high Area Under the Curve (AUC) values, indicating excellent discrimination among breast cancer molecular subtypes. Logistic Regression and Support Vector Machine consistently produced the highest ROC performance, further supporting the findings obtained from accuracy and cross-validation analyses. The ROC analysis demonstrates that the developed models can effectively distinguish between multiple molecular subtypes using gene expression data.

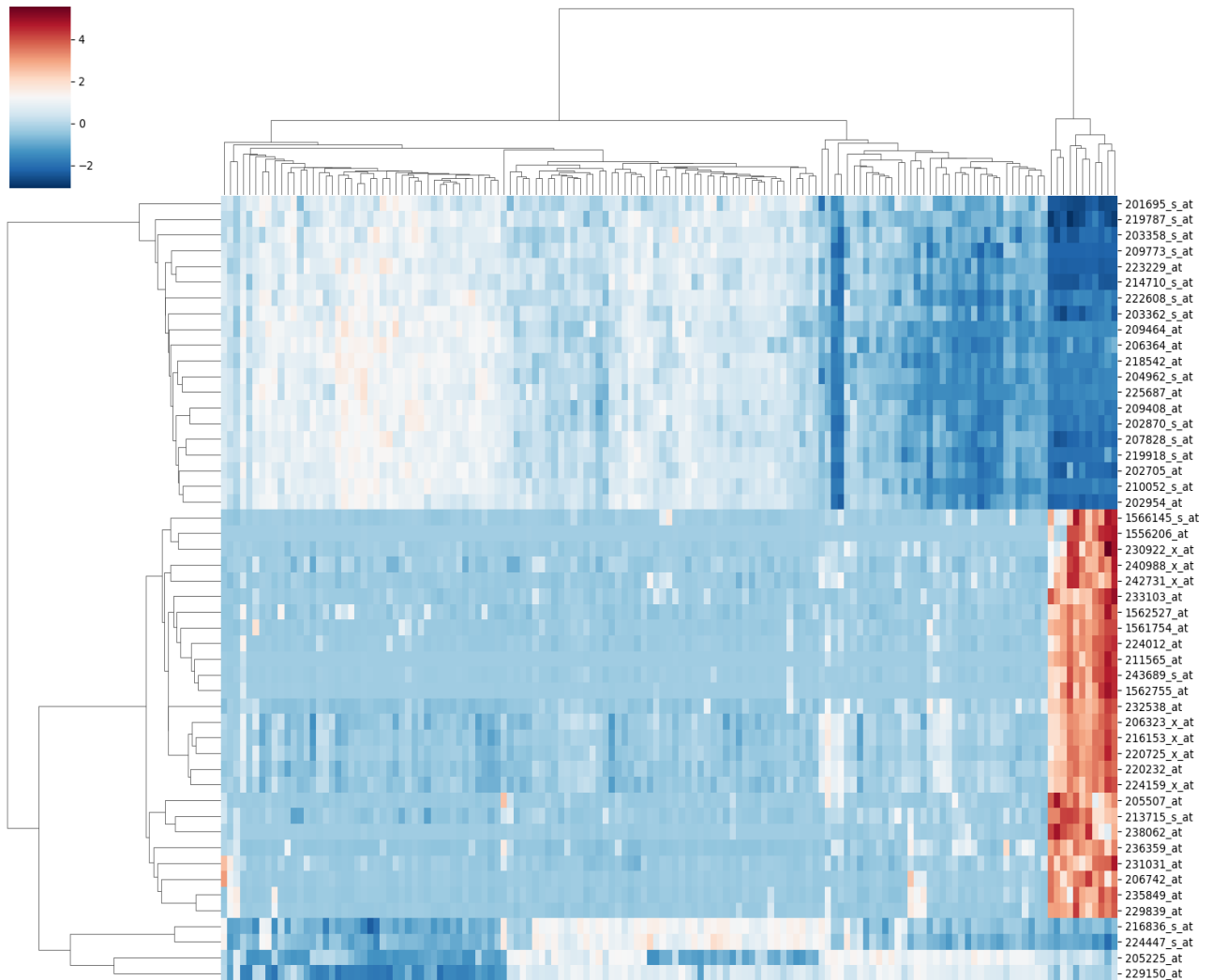
Figure 3. Receiver Operating Characteristic (ROC) curves comparing the classification performance of Logistic Regression, Support Vector Machine, Random Forest, and XGBoost. Higher AUC values indicate stronger discriminative performance.



Identification of Important Genes

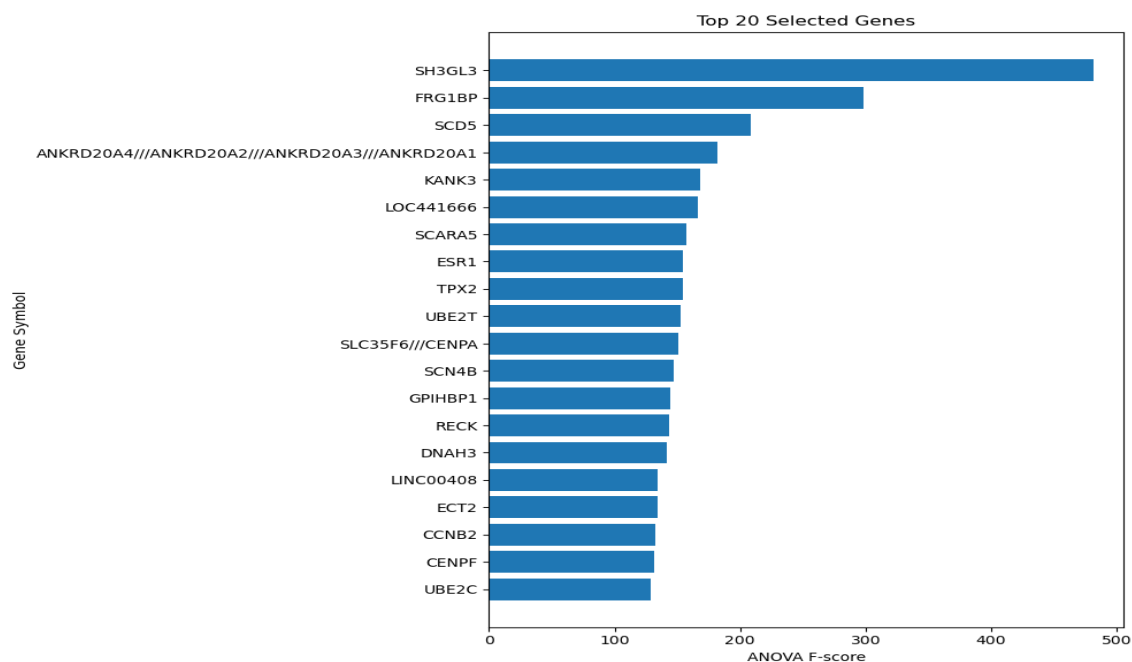
Feature selection using the ANOVA F-test identified several highly informative genes that contributed to breast cancer molecular subtype classification. Among the top-ranked genes were **ESR1, TPX2, UBE2T, CENPF, CCNB2, UBE2C, RECK, SCARA5, and KANK3**. Many of these genes have previously been reported to participate in hormone receptor signaling, cell cycle regulation, chromosome segregation, and tumor progression. The identification of these biologically relevant genes suggests that the feature selection approach successfully retained important molecular markers while reducing the dimensionality of the dataset. These genes therefore contributed not only to improved classification performance but also to the biological interpretation of the results.

Figure 4. Heatmap showing the expression patterns of the top selected genes across breast cancer molecular subtypes. Distinct expression profiles can be observed among different subtype groups.



The heatmap demonstrates distinct expression patterns across the selected genes, with several genes showing subtype-specific expression profiles. These expression differences further support the ability of the selected genes to discriminate among breast cancer molecular subtypes and highlight their potential biological relevance.

Figure 5. Top-ranked genes identified using the ANOVA F-test based on their F-score values. Higher F-scores indicate greater importance for distinguishing breast cancer molecular subtypes.



The ANOVA F-score ranking highlights genes with the greatest discriminatory power. Genes such as **ESR1**, **TPX2**, **UBE2T**, and **CCNB2** received the highest scores, suggesting that variations in their expression contribute substantially to subtype classification.

Gene Ontology (GO) Enrichment Analysis

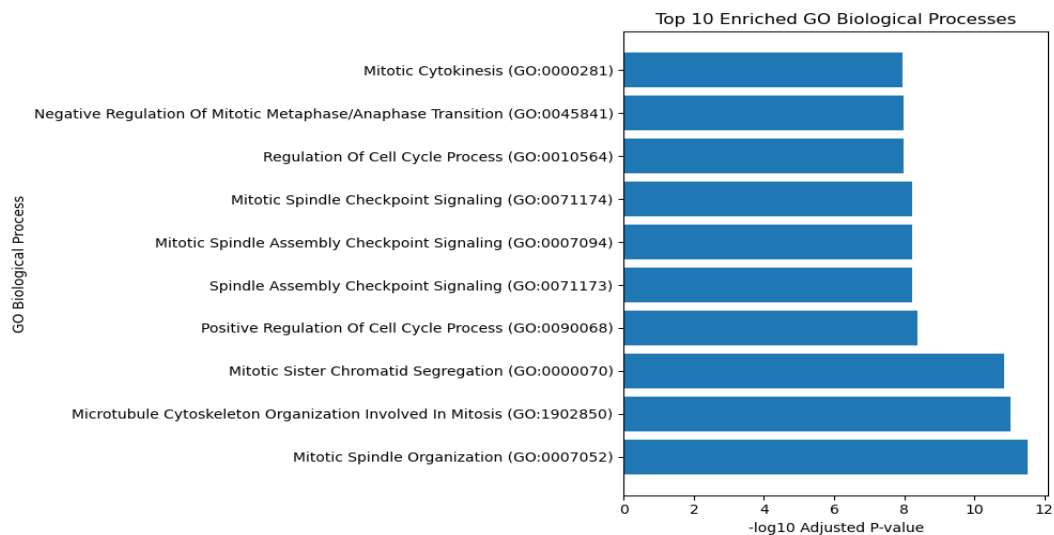
Gene Ontology (GO) enrichment analysis was carried out to understand the biological functions of the genes selected during feature selection. The analysis showed that most of the enriched biological processes were related to cell cycle regulation and mitosis, suggesting that these genes play an important role in cell division and the growth of breast cancer cells.

The most significantly enriched GO term was Mitotic Spindle Organization, followed by Microtubule Cytoskeleton Organization Involved in Mitosis and Mitotic Sister Chromatid Segregation. Other important biological processes included Positive Regulation of Cell Cycle Process, Spindle Assembly

Checkpoint Signaling, Mitotic Spindle Checkpoint Signaling, Regulation of Cell Cycle Process, Negative Regulation of Mitotic Metaphase/Anaphase Transition, and Mitotic Cytokinesis.

These processes are essential for ensuring that cells divide correctly and maintain the proper distribution of chromosomes during cell division. When these mechanisms become disrupted, uncontrolled cell growth and genomic instability can occur, which are well-known characteristics of cancer. Therefore, the GO enrichment results support the biological importance of the genes identified in this study and indicate that they are closely associated with breast cancer progression.

Figure 6. Gene Ontology (GO) enrichment analysis of the selected genes. The enriched biological processes are primarily associated with cell cycle regulation, chromosome segregation, and mitotic activity, highlighting the biological relevance of the selected features.



KEGG Pathway Enrichment Analysis

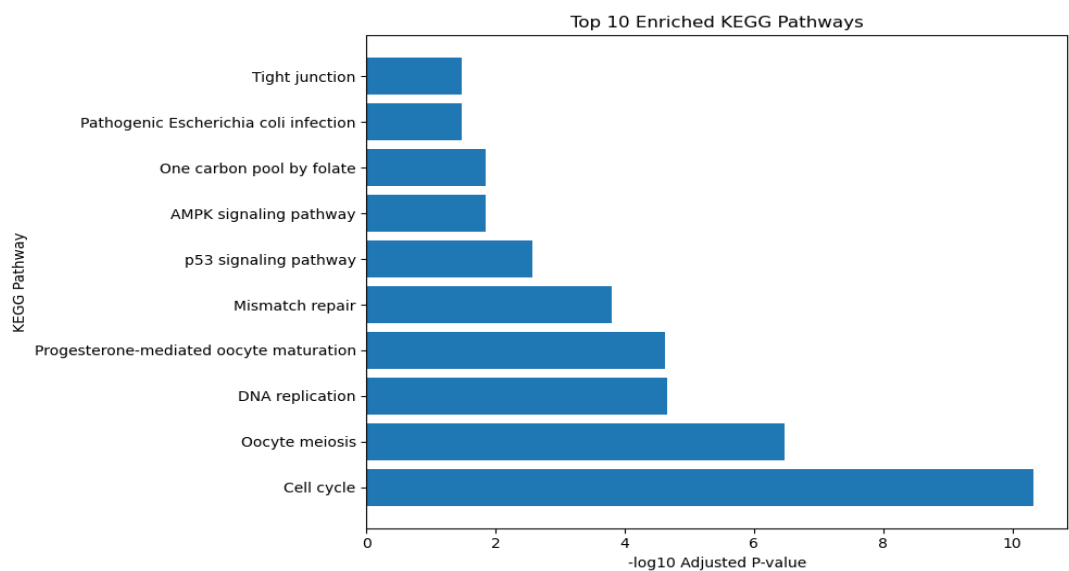
KEGG pathway enrichment analysis was performed to identify the biological pathways in which the selected genes are involved. The results showed that the genes were significantly enriched in several

pathways that are known to be associated with cell growth, DNA maintenance, and cancer development.

Among all the pathways, the Cell Cycle pathway showed the highest level of enrichment. Other significantly enriched pathways included Oocyte Meiosis, DNA Replication, Progesterone-Mediated Oocyte Maturation, Mismatch Repair, and the p53 Signaling Pathway. Additional pathways such as the AMPK Signaling Pathway, One Carbon Pool by Folate, Pathogenic *Escherichia coli* Infection, and Tight Junction were also identified.

Many of these pathways are directly involved in regulating cell division, repairing DNA damage, and maintaining normal cellular function. In particular, the enrichment of the Cell Cycle, DNA Replication, Mismatch Repair, and p53 Signaling pathways highlights their important role in breast cancer biology. These findings suggest that the genes selected by the machine learning models are not only useful for classification but are also biologically relevant and involved in key mechanisms underlying breast cancer.

Figure 7. Top enriched KEGG pathways identified from the selected genes.



Overall, both the GO and KEGG enrichment analyses consistently indicated that the selected genes are mainly involved in cell cycle regulation, mitotic processes, DNA replication, and DNA repair. These

findings support the machine learning results and demonstrate that the identified genes are biologically relevant to breast cancer molecular subtype classification.

Discussion

The primary objective of this study was to classify breast cancer molecular subtypes using gene expression data and compare the performance of different machine learning algorithms. The results demonstrated that all four machine learning models were able to classify the molecular subtypes with high accuracy, highlighting the effectiveness of machine learning techniques for analyzing high-dimensional gene expression data.

Among the evaluated models, Logistic Regression and Support Vector Machine (SVM) achieved a test accuracy of 100%. While these results are highly encouraging, test accuracy alone may not always reflect how well a model performs on unseen data, particularly when working with relatively small datasets. To obtain a more reliable assessment, 10-fold stratified cross-validation was performed. The cross-validation results showed that SVM achieved the highest mean accuracy (96.44%) with the lowest standard deviation, indicating that it provided the most stable and consistent performance across different data splits. Logistic Regression also demonstrated excellent performance, whereas Random Forest and XGBoost achieved slightly lower but still competitive accuracies. These findings suggest that carefully selected machine learning models can effectively distinguish breast cancer molecular subtypes using gene expression profiles.

One of the key components of this study was the application of the ANOVA F-test for feature selection. Gene expression datasets typically contain thousands of genes, many of which contribute little to disease classification. By selecting only the most informative genes, the dimensionality of the dataset was substantially reduced while preserving the features most relevant for subtype classification. This not

only improved computational efficiency but also allowed the study to focus on genes with greater biological significance, making the predictive models easier to interpret.

To further investigate the biological relevance of the selected genes, Gene Ontology (GO) and KEGG pathway enrichment analyses were performed. The GO analysis revealed that the selected genes were primarily associated with biological processes involved in mitosis, chromosome segregation, cytokinesis, and cell cycle regulation. Similarly, the KEGG analysis identified significant enrichment in pathways such as the Cell Cycle, DNA Replication, Mismatch Repair, and p53 Signaling Pathway. These pathways play essential roles in regulating cell growth, maintaining genomic stability, and preventing uncontrolled cell division, all of which are closely associated with breast cancer development and progression. The agreement between the GO and KEGG analyses further supports the biological relevance of the genes identified during feature selection and demonstrates that the computational findings are consistent with well-established mechanisms of breast cancer biology.

Comparison with Previous Studies

The findings of this study are consistent with previous research demonstrating the effectiveness of machine learning for breast cancer subtype classification using gene expression data. Earlier studies have reported that algorithms such as Support Vector Machine, Random Forest, and XGBoost can achieve high classification performance when combined with appropriate feature selection techniques. Similar to these studies, the present work achieved excellent predictive performance, particularly with Logistic Regression and SVM. However, unlike many previous studies that primarily focused on classification accuracy, this research also incorporated functional enrichment analysis through GO and KEGG pathways. By combining predictive modeling with biological interpretation, this study provides a more comprehensive understanding of the molecular mechanisms underlying breast cancer subtype classification rather than focusing solely on model performance.

Strengths of the Study

One of the main strengths of this study is the integration of both computational and biological analyses within a single workflow. Instead of evaluating only one machine learning model, four widely used algorithms were systematically compared to identify the most effective classifier. The use of 10-fold stratified cross-validation provided a more reliable assessment of model performance and reduced the possibility of reporting overly optimistic results based on a single train-test split. Furthermore, incorporating gene annotation, heatmap visualization, and GO and KEGG enrichment analyses added biological context to the selected genes, making the findings more meaningful and easier to interpret from a biomedical perspective.

Limitations of the Study

Despite the encouraging results, this study has several limitations. The analysis was conducted using a single publicly available microarray dataset containing 141 samples after preprocessing, which may limit the generalizability of the findings to broader patient populations. Although cross-validation helped improve the reliability of model evaluation, validation using independent external datasets would provide stronger evidence of model robustness. In addition, the study focused exclusively on microarray gene expression data and did not consider other types of genomic information, such as RNA sequencing, DNA methylation, or proteomics data. Experimental validation of the identified genes was also beyond the scope of this study.

Future Scope

Future research can build upon this work by validating the developed models on larger and independent breast cancer datasets, including data from resources such as TCGA and additional GEO studies. The application of deep learning approaches and ensemble learning methods may further improve classification performance. Integrating multiple omics datasets could also provide a more comprehensive understanding of breast cancer biology and help identify more robust biomarkers. Finally, experimental validation of the selected genes would strengthen their potential clinical relevance.

and support the development of precision medicine approaches for breast cancer diagnosis and treatment.

Overall, this study demonstrates that combining machine learning with statistical feature selection and functional enrichment analysis can effectively classify breast cancer molecular subtypes while also providing valuable biological insights. By integrating predictive modeling with biological interpretation, the study contributes to a deeper understanding of the molecular basis of breast cancer and highlights the growing potential of computational approaches to support cancer research and advance precision medicine.

Novelty of the Study

The novelty of this study lies in the integration of statistical feature selection, comparative evaluation of multiple machine learning algorithms, and biological interpretation through GO and KEGG enrichment analysis using the GSE45827 dataset. Rather than focusing solely on predictive accuracy, the study combines computational modeling with functional analysis to provide insights into the molecular mechanisms underlying breast cancer subtype classification. This integrated approach enhances both the predictive and biological relevance of the findings.

Conclusion

This study demonstrated the application of machine learning techniques for the classification of breast cancer molecular subtypes using gene expression data from the GSE45827 dataset. After preprocessing the data and selecting the most informative genes using the ANOVA F-test, four machine learning algorithms—Logistic Regression, Support Vector Machine (SVM), Random Forest, and XGBoost—were trained and evaluated.

Among the evaluated models, Logistic Regression and SVM achieved the highest test accuracy of 100%, while cross-validation results showed that SVM provided the most consistent overall performance. These findings indicate that machine learning models are capable of accurately distinguishing between different breast cancer molecular subtypes using gene expression profiles.

The biological interpretation of the selected genes further strengthened the study. GO enrichment analysis revealed that many of the identified genes are involved in cell cycle regulation and mitotic processes, while KEGG pathway analysis highlighted important pathways such as the Cell Cycle, DNA Replication, Mismatch Repair, and p53 Signaling Pathway. These findings are consistent with the known molecular mechanisms involved in breast cancer development and progression.

Although the study was conducted using a relatively small publicly available dataset, the combination of feature selection, cross-validation, and functional enrichment analysis improved the reliability and biological relevance of the results. Future work may include validation using independent datasets, RNA sequencing data, and deep learning approaches to further improve prediction performance.

In conclusion, this study shows that integrating machine learning with biological pathway analysis provides an effective approach for breast cancer molecular subtype classification. The findings contribute to a better understanding of breast cancer biology and demonstrate the growing potential of

integrating machine learning and bioinformatics to support breast cancer research and advance future precision medicine.

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